RAG for CV Papers

Problem and motivation

- AI technology is advancing rapidly. Even when focusing only on the field of computer vision, many new papers are published every day.
- Reviewing the original papers is time-consuming, and most are not relevant.

Solution

- Automatically fetch the latest trends from hugging face and download the corresponding papers from arXiv every day.
- Summarize papers within 7 days by a single click, while a vector database helps prevent hallucinations.

Data used : Trend on hugging face, Papers from arXiv.

Specific queries

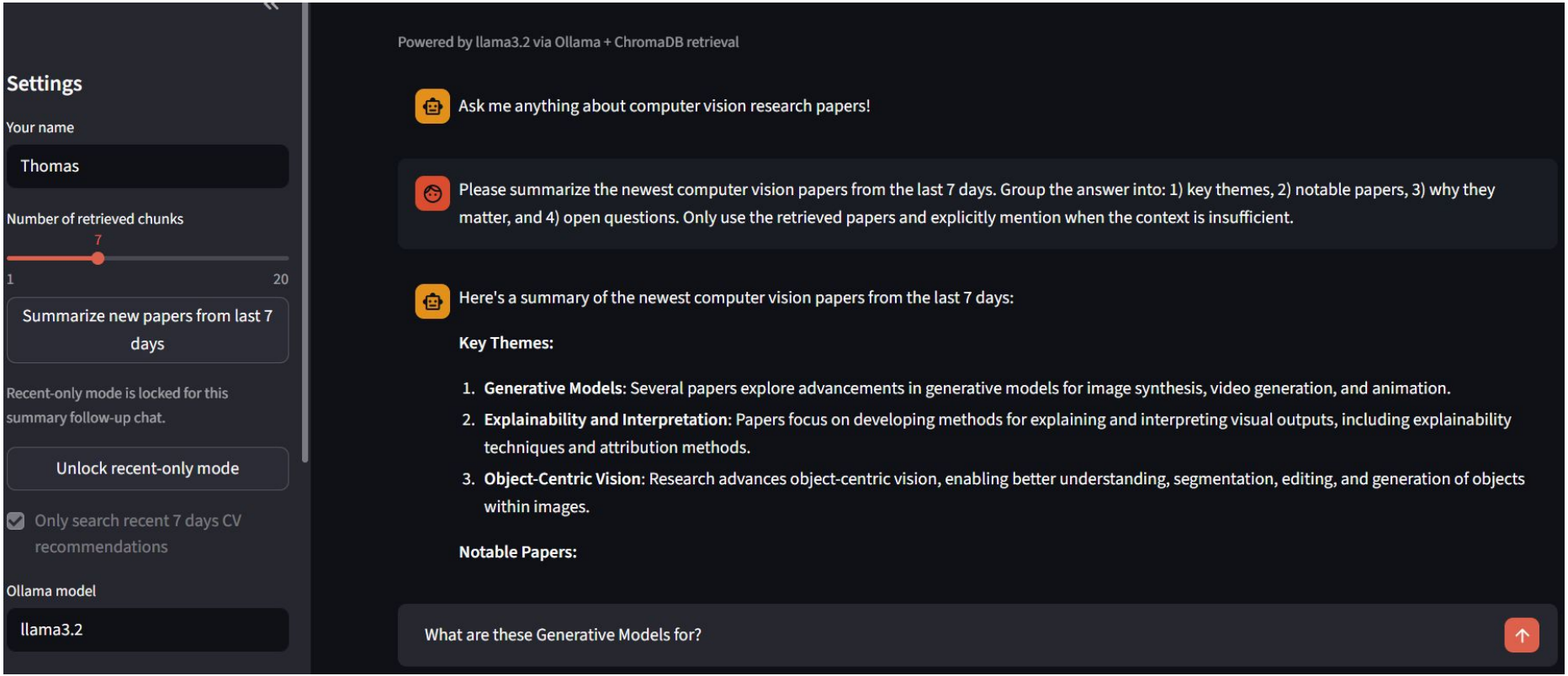
- Click the button to receive a summary.
- Ask following question from the summary.
- Related topic of “combine diffusion models with reinforcement learning ”

Live Endpoint

Repository

Technical Writeup

Summary


Settings
Your name
Thomas
Number of retrieved chunks
7
1
20
Summarize new papers from last 7 days
Recent-only mode is locked for this summary follow-up chat.
Unlock recent-only mode
Only search recent 7 days CV recommendations
Ollama model
llama3.2
Powered by llama3.2 via Ollama + ChromaDB retrieval
Ask me anything about computer vision research papers!
Please summarize the newest computer vision papers from the last 7 days. Group the answer into: 1) key themes, 2) notable papers, 3) why they matter, and 4) open questions. Only use the retrieved papers and explicitly mention when the context is insufficient.
Here's a summary of the newest computer vision papers from the last 7 days:
Key Themes:
1. Generative Models: Several papers explore advancements in generative models for image synthesis, video generation, and animation.
2. Explainability and Interpretation: Papers focus on developing methods for explaining and interpreting visual outputs, including explainability techniques and attribution methods.
3. Object-Centric Vision: Research advances object-centric vision, enabling better understanding, segmentation, editing, and generation of objects within images.
Notable Papers:
What are these Generative Models for?

Summary

What are these Generative Models for?

Key Themes:

- Generative modeling in computer vision, particularly in image and video generation
- Improving scalability, diversity, and quality of generated samples
- Development of new architectures and techniques to address the generative learning trilemma (sampling speed, diversity, and quality)
- Application of generative models to various tasks such as image restoration, face detection, and scene generation

Notable Papers:

- [1] BlendFusion: Scalable Synthetic Data Generation for Diffusion Model Training
  - This paper focuses on generating synthetic data for training diffusion models, which can improve the quality and diversity of generated samples.
- [2] MixFlow: Mixed Source Distributions Improve Rectified Flows
  - This paper proposes a new approach to mixing source distributions in rectified flows, aiming to improve the performance of generative models while reducing computational costs.
- [3] ELT: Elastic Looped Transformers for Visual Generation
  - This paper introduces an elastic looped transformer architecture for visual generation, which can generate high-quality samples with improved